



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)

First Semester M.Tech (MMD) (CBCS) Degree Examinations

December 2024 - January 2025

22MMD113 – MECHANISM DESIGN

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I

Marks BT* CO* PO*

1. a) Define the following:

- i. Mechanism
- ii. Machine
- iii. Rigid Body
- iv. Kinematic Pair
- v. Degrees of Freedom

10 L*1 1 1

b) Illustrate Grubler's Criterion. Calculate DOF for the following kinematic linkage.

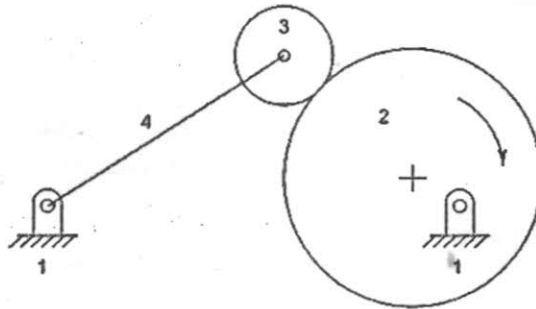


Fig. Q1(b)

10 L2 1 2

2. a) Illustrate kinematic pairs according to the nature of mechanical constraint.

10 L2 1 1

b) Find the maximum and minimum transmission angles for the mechanism shown in Figure Q2(b).

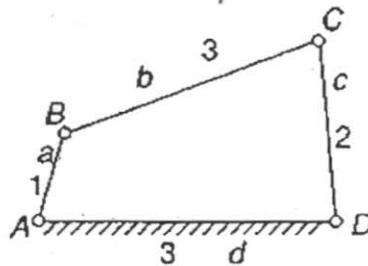


Fig. Q2(b)

10 L3 2 2

3. a) Figure Q3(a) shows the configuration of an engine mechanism. The dimensions are the following:

2

Crank OA = 20mm, Connecting rod AB = 600mm, distance of center of mass from crank end AD = 200mm. At the instant, the crank has an angular velocity of 50 rad/s clockwise and an angular acceleration of 800 rad/s². Calculate the (i) velocity of D and angular velocity of AB (ii) acceleration of D and angular acceleration of AB.

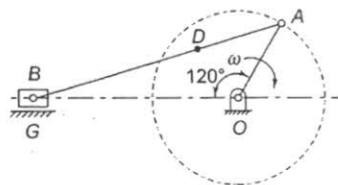


Fig. Q3(a)

20 L3 2

Unit – II

4. a) Define the following:
- Kinematic synthesis
 - Number and dimensional synthesis
 - Function generation
 - Path generation
 - Body guidance
- b) Briefly explain the two-position synthesis method of a slider-crank mechanism with a neat diagram.
5. a) Briefly explain the two-position synthesis method of a crank-rocker mechanism with a neat diagram.
- b) Design a four-bar mechanism governed by the equation $Y=X^{1.5}$ and X varies from 1 to 4. Assume θ to differ from 30° to 120° and ϕ from 60° to 130° . The length of the fixed link is 30mm. Use Chebyshev spacing of accuracy points. Analyze using Grashoff's criterion.
6. a) Design a four-bar mechanism using Freudenstein's equation when a function $Y=2X^2$ governs the motions of the input and output links and X varies from 2 to 4 with an interval of 1. Assume θ to differ from 40° to 120° and ϕ from 60° to 132° . Analyze using Grashoff's criterion.

10	L1	3	1
10	L2	3	1
10	L2	3	1
10	L4	4	2
20	L4	4	2

Unit – III

7. a) Determine the torque exerted by the crankshaft on AB for the four-bar mechanism shown in Fig. Q7(a), given that T_3 on link BC is 30 Nm clockwise and T_4 on CD is 20 Nm counter-clockwise. The link dimensions are AD = 800 mm, AB = 300 mm, BC = 700 mm and CD = 400 mm.

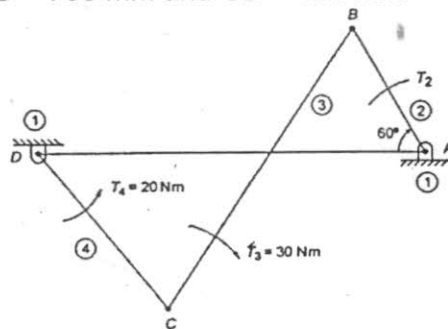


Fig. Q 7(a)

10	L3	5	2
10	L3	5	2

- b) A horizontal gas engine running at 210 rpm has a bore of 220 mm and a stroke of 440 mm. The connecting rod is 924 mm long and the reciprocating parts weigh 20 kg. When the crank has rotated through an angle of 30° from the inner dead center, the gas pressures on the cover and crank sides are 500 kN/m^2 and 60 kN/m^2 respectively. The diameter of the piston rod is 40 mm. Determine (i) the turning moment on the crankshaft, (ii) the thrust on the bearings, and (iii) the acceleration of the flywheel which has a mass of 8 kg and radius of gyration of 600 mm while the power of the engine is 22 kW.

8. The following list of data corresponds to the reciprocating engine mechanism shown in Figure Q8(a):
 Length of crank = 7.5 cm, Length of connecting rod = 28 cm, Distance of centre of gravity (C.G.) of link 2 from main bearings = 5 cm, Distance of centre of gravity of link 3 from crank pin = 12 cm, Crank angle from line of stroke = 60° , Crank speed = 2000 rpm counter-clockwise, Mass of link 2 = 2.5 kg, Mass of link 3 = 4 kg, Mass of link 4 = 3 kg, Mass moment of inertia of link 2 = 60 kgcm^2 and mass moment of inertia of link 3 = 500 kgcm^2 . Compute a complete inertia force analysis for the mechanism.

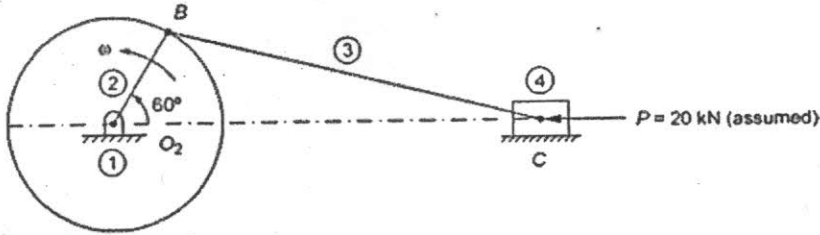


Fig.Q8(a)

20 L3 5 2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
